Magnetic Effects of Electric Current

The big idea

An electric current produces a **magnetic field**. Hans Christian Oersted noticed that a compass needle deflects near a current-carrying wire. This link between electricity and magnetism runs motors, loudspeakers and MRI machines.

Magnetic field and field lines

A **magnetic field** exists around a magnet. **Field lines** show it: they leave the **north pole** and enter the **south pole** outside the magnet, and go from south to north inside it. They are **closed curves**, **never cross** (the field can't point two ways at once), and are **closer together where the field is stronger**.

Field due to a current

Conductor	Shape of field	Notes
Straight wire	Concentric circles around the wire	Stronger near the wire and for larger current
Circular loop	Lines nearly straight at the centre	Field increases with the number of turns
Solenoid (long coil)	Like a bar magnet; **uniform inside**	Used to make electromagnets with a soft iron core

RIGHT-HAND THUMB RULE

Hold the wire in your right hand with the thumb pointing along the current. Your **curled fingers** show the direction of the magnetic field lines.

Force on a current-carrying conductor

A conductor carrying current in a magnetic field experiences a **force**. It is maximum when the current is **perpendicular** to the field.

FLEMING'S LEFT-HAND RULE

Stretch the thumb, forefinger and middle finger of your left hand mutually perpendicular. **F**orefinger = **F**ield, **c**entre (middle) finger = **c**urrent, **th**umb = **th**rust (force/motion).

This is the principle of the **electric motor** and loudspeakers. In medicine, **MRI** uses magnetic fields produced inside the body to form images. Even nerve impulses create weak magnetic fields.

Domestic electric circuits

In India, the supply is **220 V AC at 50 Hz**.

THREE WIRES

Live (red insulation), **neutral** (black), **earth** (green). Earth wire connects metal bodies of appliances to the ground so that leaked current flows to earth instead of giving you a shock.

- Separate circuits: **15 A** for high-power appliances (geysers, ACs), **5 A** for bulbs and fans.
- Appliances are in **parallel**.
- **Short circuit**: live and neutral touch, current rises sharply.
- **Overloading**: too many appliances on one socket, or a voltage surge.
- The **fuse** (or MCB) protects against both.

Quick recap

- Current creates a magnetic field; right-hand thumb rule gives its direction.
- A solenoid's field is uniform inside, like a bar magnet.
- Fleming's left-hand rule gives the force on a conductor.
- Household wiring: live, neutral, earth; fuses protect against short circuits and overloading.

Practice questions

1. Magnetic field lines never intersect because

   (a) the field would have two directions at one point (b) they are parallel (c) they are closed curves (d) the field is uniform

2. The magnetic field inside a long solenoid is

   (a) zero (b) uniform (c) strongest at the ends only (d) circular

3. Fleming's left-hand rule gives the direction of the

   (a) induced current (b) magnetic field (c) force on a current-carrying conductor (d) electric field

4. The colour of the earth wire is usually

   (a) red (b) black (c) green (d) blue

5. The domestic supply in India is

   (a) 110 V, 60 Hz (b) 220 V, 50 Hz (c) 220 V, 60 Hz (d) 440 V, 50 Hz

Answers: 1. a 2. b 3. c 4. c 5. b